

Solutions

Exercise 1

First, let's calculate the net input for each of the perceptrons using the formula:

$$z = w \cdot x + b$$

$$z_1 = w_1 \cdot x + b_1 = (0.5 \times 2 + 1.5 \times (-1)) - 0.5 = -1$$

$$z_2 = w_2 \cdot x + b_2 = (-2 \times 2 + 0.5 \times (-1)) + 1 = -3.5$$

Second, we apply the corresponding activation function to the output.

1. Heaviside Step Function

$$H(z) = \begin{cases} 1, & z \geq 0 \\ 0, & z < 0 \end{cases}$$

$$H(z_1) = H(-1) = 0$$

$$H(z_2) = H(-3.5) = 0$$

2. ReLU (Rectified Linear Unit)

$$\text{ReLU}(z) = \max(0, z)$$

$$\text{ReLU}(z_1) = \max(0, -1) = 0$$

$$\text{ReLU}(z_2) = \max(0, -3.5) = 0$$

3. Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\sigma(z_1) = \frac{1}{1 + e^1} \approx 0.2689$$

$$\sigma(z_2) = \frac{1}{1 + e^{3.5}} \approx 0.0293$$

4. Tanh Function

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\tanh(z_1) = \frac{e^{-1} - e^1}{e^{-1} + e^1} \approx -0.7616$$

$$\tanh(z_2) = \frac{e^{-3.5} - e^{3.5}}{e^{-3.5} + e^{3.5}} \approx -0.9982$$

Exercise 2

First, we choose a possible linear decision boundary that separates the dark points (class 1) from the white points (class 0).

Next, we determine the weights and biases. The weight vectors should be perpendicular to the decision boundaries and directed toward the points classified as 1 (the dark points). Their length can be chosen freely. Here is one possible selection for the weight vectors:

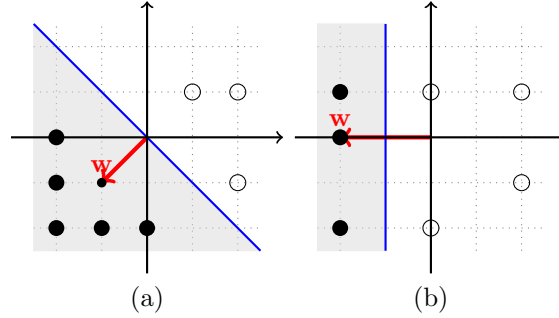


Figure 1: Decision boundary

Here is one set of choices for the weight vectors:

$$\begin{aligned} \text{(a)} \quad w_1^T &= [-1 \quad -1], \\ \text{(b)} \quad w_2^T &= [-2 \quad 0] \end{aligned}$$

Now we find the bias values for each perceptron by picking a point on the decision boundary and using the boundary condition $w^T \times p + b = 0 \Leftrightarrow -w^T \times p = b$. This gives us the following biases:

$$\begin{aligned} \text{(a)} \quad b_1 &= -[-1 \quad -1] \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0 \\ \text{(b)} \quad b_2 &= -[-2 \quad 0] \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -2 \end{aligned}$$

Exercise 3

Iteration 1: $(x_1, x_2) = (1, 1)$, $y = 1$:

$$O_1 = \text{sgn}(w_1 \cdot 1 + w_2 \cdot 1 + b) = \text{sgn}(0) = 0$$

$$w_1 \leftarrow w_1 + 0.1 \cdot (1 - 0) \cdot 1 = 0 + 0.1 = 0.1$$

$$w_2 \leftarrow w_2 + 0.1 \cdot (1 - 0) \cdot 1 = 0 + 0.1 = 0.1$$

$$b \leftarrow b + 0.1 \cdot (1 - 0) = 0 + 0.1 = 0.1$$

Iteration 2: $(x_1, x_2) = (1, -1)$, $y = -1$:

$$O_2 = \text{sgn}(w_1 \cdot 1 + w_2 \cdot (-1) + b) = \text{sgn}(0.1 \cdot 1 + 0.1 \cdot (-1) + 0.1) = \text{sgn}(0.1) = 1$$

$$w_1 \leftarrow w_1 + 0.1 \cdot (-1 - 1) \cdot 1 = 0.1 + (-0.2) = -0.1$$

$$w_2 \leftarrow w_2 + 0.1 \cdot (-1 - 1) \cdot (-1) = 0.1 + 0.2 = 0.3$$

$$b \leftarrow b + 0.1 \cdot (-1 - 1) = 0.1 + (-0.2) = -0.1$$

Iteration 3: $(x_1, x_2) = (-1, 1)$, $y = -1$:

$$O_3 = \text{sgn}(w_1 \cdot (-1) + w_2 \cdot 1 + b) = \text{sgn}((-0.1) \cdot (-1) + 0.3 \cdot 1 + (-0.1)) = \text{sgn}(0.3) = 1$$

$$w_1 \leftarrow w_1 + 0.1 \cdot (-1 - 1) \cdot (-1) = -0.1 + 0.2 = 0.1$$

$$w_2 \leftarrow w_2 + 0.1 \cdot (-1 - 1) \cdot 1 = 0.3 + (-0.2) = 0.1$$

$$b \leftarrow b + 0.1 \cdot (-1 - 1) = -0.1 + (-0.2) = -0.3$$

Iteration 4: $(x_1, x_2) = (-1, -1)$, $y = -1$:

$$O_4 = \text{sgn}(w_1 \cdot (-1) + w_2 \cdot (-1) + b) = \text{sgn}(0.1 \cdot (-1) + 0.1 \cdot (-1) + (-0.3)) = \text{sgn}(-0.5) = -1$$

Since $O_4 = -1$, no update is needed.

Thus, after one full iteration of weight updates, the final weights and bias are:

$$w_1 = 0.1, \quad w_2 = 0.1, \quad b = -0.3$$

Exercise 4

$$x = \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix}, \quad y = 0.6,$$

$$W^1 = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}, \quad b^1 = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix},$$

$$W^2 = \begin{bmatrix} 0.5 & 0.6 \end{bmatrix}, \quad b^2 = 0.1,$$

Learning rate: $\eta = 0.1$

1. Forward Pass

Net input from the first layer :

$$z_1 = W^1 x + b^1 = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 0.3 \\ 0.7 \end{bmatrix} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} = \begin{bmatrix} 0.27 \\ 0.47 \end{bmatrix}$$

ReLU activation:

$$O_1 = \text{ReLU}(z_1) = \begin{bmatrix} \max(0, 0.27) \\ \max(0, 0.47) \end{bmatrix} = \begin{bmatrix} 0.27 \\ 0.47 \end{bmatrix}.$$

Net input from the second layer :

$$z_2 = W^2 O_1 + b^2 = \begin{bmatrix} 0.5 & 0.6 \end{bmatrix} \begin{bmatrix} 0.27 \\ 0.47 \end{bmatrix} + 0.1 = 0.517$$

The sigmoid activation:

$$O_2 = \sigma(z_2) = \frac{1}{1 + e^{-0.517}} \approx 0.626.$$

2. Loss Calculation

For a two-dimensional input the MSE is calculated as

$$L = \frac{1}{2}(O_2 - y)^2 = \frac{1}{2}(0.626 - 0.6)^2 \approx 0.000338.$$

3. Backpropagation

First, compute the derivative of the loss with respect to O_2 :

$$\frac{\partial L}{\partial O_2} = O_2 - y = 0.626 - 0.6 = 0.026.$$

The derivative of the sigmoid is:

$$\sigma'(z_2) = \sigma(z_2)(1 - \sigma(z_2)) = O_2(1 - O_2) \approx 0.626 \times 0.374 \approx 0.234.$$

Thus, the error term at the output is:

$$\delta_2 = \frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial O_2} \sigma'(z_2) \approx 0.026 \times 0.234 \approx 0.006.$$

Now, let's compute the gradients for W^2 :

$$\frac{\partial L}{\partial W^2} = \delta_2 \cdot O_1^T = 0.006 \begin{bmatrix} 0.27 & 0.47 \end{bmatrix} \approx \begin{bmatrix} 0.0016 & 0.0028 \end{bmatrix},$$

$$\frac{\partial L}{\partial b^2} = \delta_2 \approx 0.006.$$

Hidden Layer: First, let's calculate the error term δ_1 :

For the hidden layer with ReLU, both components of z_1 are positive, so

$$\text{ReLU}'(z_1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\delta_1 = (W^{2\top} \delta_2) \odot \text{ReLU}'(z_1) = W^{2\top} \delta_2 = \begin{bmatrix} 0.5 \\ 0.6 \end{bmatrix} \cdot 0.006 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 0.0030 \\ 0.0036 \end{bmatrix}$$

Thus, the gradients for W^1 and b^1 are:

$$\frac{\partial L}{\partial W^1} = \delta_1 \cdot x^T = \begin{bmatrix} 0.0030 \\ 0.0036 \end{bmatrix} \begin{bmatrix} 0.3 & 0.7 \end{bmatrix} \approx \begin{bmatrix} 0.0009 & 0.0021 \\ 0.001 & 0.0025 \end{bmatrix},$$

$$\frac{\partial L}{\partial b^1} = \delta_1 \approx \begin{bmatrix} 0.003 \\ 0.0036 \end{bmatrix}.$$

4. Parameter Updates

Using the update rule

$$\theta_{\text{new}} = \theta_{\text{old}} - \eta \frac{\partial L}{\partial \theta},$$

we have:

$$W^{2 \text{ new}} = W^2 - 0.1 \cdot \frac{\partial L}{\partial W^2} \approx \begin{bmatrix} 0.5 & 0.6 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 0.0016 & 0.0028 \end{bmatrix} \approx \begin{bmatrix} 0.49984 & 0.59972 \end{bmatrix},$$

$$b^{2 \text{ new}} = b^2 - 0.1 \cdot \frac{\partial L}{\partial b^2} \approx 0.1 - 0.1 \cdot 0.006 \approx 0.0994.$$

$$W^{1 \text{ new}} = W^1 - 0.1 \cdot \frac{\partial L}{\partial W^1} \approx \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 0.0009 & 0.0021 \\ 0.001 & 0.0025 \end{bmatrix} \approx \begin{bmatrix} 0.09991 & 0.19979 \\ 0.2999 & 0.39975 \end{bmatrix}$$

$$b^{1 \text{ new}} = b^1 - 0.1 \cdot \frac{\partial L}{\partial b^1} \approx \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix} - 0.1 \cdot \begin{bmatrix} 0.003 \\ 0.0036 \end{bmatrix} \approx \begin{bmatrix} 0.0997 \\ 0.09964 \end{bmatrix}.$$

Exercise 5

- (a) OR: $w_1 = 2 \quad w_2 = 2 \quad b = -1$
- (b) NOT: $w_1 = -1 \quad b = 1$
- (c) NOR: $w_1 = -1 \quad w_2 = -1 \quad b = 1$
- (d) NAND: $w_1 = -1 \quad w_2 = -1 \quad b = 2$
- (e) XOR:

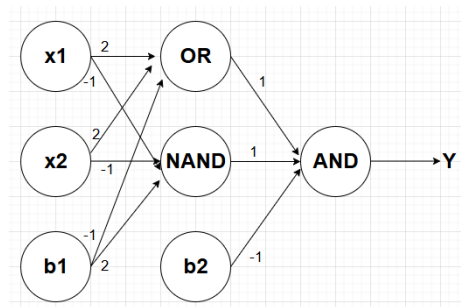


Figure 2: XOR